Sequences & Patterns

Section 1: Arithmetic Sequences & Constant Differences

1 What is the next number in the arithmetic sequence: $4, 11, 18, 25, \ldots$?

a. 32

b. 31

c. 33

d. 35

2 In an arithmetic sequence where the first term is 3 and the common difference is 4, what is the 15th term?

a. 59

b. 63

c. 55

d. 60

3 What is the sum of the first 20 positive even integers $(2 + 4 + 6 + \ldots + 40)$?

a. 420

b. 400

c. 440

d. 380

Section 2: Geometric & Exponential Growth Sequences

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 What is the next term in the geometric sequence: $3, 6, 12, 24, \ldots$?

a. 48

b. 36

c. 42

d. 54

5 A bacterial culture doubles in number every 20 minutes. If there are initially 100 bacteria, how many will there be after 2 hours?

a. 6400

b. 3200

c. 12800

d. 1600

6 What is the 6th term of the sequence: $2, -6, 18, -54, \ldots$?

a. -486

b. 486

c. -162

d. 162

Section 3: Composite, Alternating & Two-Step Patterns

7 Find the missing number in the sequence: $2, 3, 5, 8, 13, 21, \ldots$?

a. 34

b. 32

c. 33

d. 35

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 What is the next number in the pattern: $1, 4, 9, 16, 25, \ldots$?

a. 36

b. 35

c. 40

d. 49

9 Find the next term in the sequence: $2, 5, 11, 23, 47, \ldots$?

a. 95

b. 94

c. 91

d. 89

Section 4: Periodic Cycles & Calendar Patterns

10 If today is Tuesday, what day of the week will it be 100 days from now?

a. Thursday

b. Wednesday

c. Friday

d. Tuesday

11 A light flashes every 12 seconds, and a second light flashes every 18 seconds. If they flash together right now, after how many seconds will they flash together again?

a. 36 seconds

b. 24 seconds

c. 48 seconds

d. 72 seconds

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 In a repeating decimal $0.\overline{142857}$, what is the 50th digit after the decimal point?

a. 8

b. 4

c. 2

d. 5

© ClipSAT: By Mr. Mohamed Abdallah +966597688647